

Research Interest

- **Multimodal Learning** inspired by human cognition for **Multimodal Agentic AI**, especially **vision-language models** and **spatial understanding**.
- My line of work in "Language for 3D Vision" explores how vision-language models can perceive and understand the world like humans do.

Educations

Yale University

Aug. 2023 – [Expected] May. 2027

-Ph.D. in Computer Science, advised by Prof. Yuval Kluger (primary) and Prof. Daniel Rakita

ShanghaiTech University

Sept. 2019 - June 2023

-B.Eng. in Computer Science, Major | Innovation and Entrepreneurship, Minor

- Entrepreneurial Mindshift Program, Babson College, March 2021 | International Summer Program, Osaka University, June 2022

Publications (First Author / Co-first Author) (*: equal contribution)

- **Iris: Integrating Language into Diffusion-based Monocular Depth Estimation**
Ziyao Zeng*, Jingcheng Ni*, Daniel Wang, Patrick Rim, Younjoon Chung, Fengyu Yang, Byung-Woo Hong, Alex Wong
CVPR 2026 | NECV 2025 Oral Presentation (18.75%) | Link: <https://adonis-galaxy.github.io/Iris-website/>
 - Language improves diffusion-based depth estimators through the conditional distributions modelled in text-to-image pre-training.
 - Language improves depth in **ambiguous regions**, enables **iterative depth refinement**, and **speed up both training and inference**.
- **RSA: Resolving Scale Ambiguities in Monocular Depth Estimators through Language Descriptions**
Ziyao Zeng, Yangchao Wu, Hyoungseob Park, Daniel Wang, Fengyu Yang, Stefano Soatto, Dong Lao, Byung-Woo Hong, Alex Wong
NeurIPS 2024 | Link: <https://arxiv.org/abs/2410.02924>
 - We are the **first** to use **language description** to infer scales of 3D scenes.
 - Trained on multiple datasets, RSA can serve **for general relative to metric depth alignment in a zero-shot setting**.
- **WorDepth: Variational Language Prior for Monocular Depth Estimation**
Ziyao Zeng, Daniel Wang, Fengyu Yang, Hyoungseob Park, Yangchao Wu, Stefano Soatto, Byung-Woo Hong, Dong Lao, Alex Wong
CVPR 2024 | Link: <https://arxiv.org/abs/2404.03635>
 - We formulate **language as a variational prior** to resolve the scale ambiguity in monocular depth estimation.
 - We train a text-VAE generating plausible depth map given a text as prior, then use **image as condition** to select most likely depth as **posterior**.
- **Can Language Understand Depth?**
Renrui Zhang*, Ziyao Zeng*, Ziyu Guo, Yafeng Li
ACM Multimedia 2022 (Brave New Idea, 12.5%) | Link: <https://arxiv.org/abs/2207.01077>
 - We are the **first** to conduct **zero-shot training-free adaptation** from semantic language to quantified vision tasks (monocular depth estimation).
 - Our DepthCLIP surpasses existing unsupervised methods and even **approaches the early fully-supervised networks**.
- **RuleSmith: Multi-Agent LLMs for Automated Game Balancing**
Ziyao Zeng, Chen Liu, Tianyu Liu, Hao Wang, Xiaotao Sun, Fengyu Yang, Xiaofeng Liu, Zhiwen Fan
arXiv technical report, 2026 | NE Agents Day 2026 | Link: <https://adonis-galaxy.github.io/RuleSmith-website/>
 - Multi-agent LLMs self-play with Bayesian Optimization to automatically balance numerical parameters in asymmetric strategic games.
 - Demonstrate LLMs can play complex strategic games using only rule books, showing strong adaptation and generalization of RuleSmith.
- **Coffee: Controllable Diffusion Fine-tuning**
Ziyao Zeng, Jingcheng Ni, Ruyi Liu, Alex Wong
arXiv technical report, 2025 | Link: <https://arxiv.org/abs/2511.14113>
 - Coffee controls diffusion fine-tuning to **prevent learning and entangling undesired concepts**.
 - We use **language as a regularization**, enabling **training-free**, **concept-flexible**, and **architecture-agnostic** fine-tuning.
- **DSPoint: Dual-scale Point Cloud Recognition with High-frequency Fusion**
Renrui Zhang*, Ziyao Zeng*, Ziyu Guo*, Xinben Gao, Kexue Fu, Jianbo Shi
SMC 2023 | Link: <https://arxiv.org/abs/2111.10332>
 - We proposed DSPoint to **conduct dual-scale processing** with local point convolution and global voxel attention for efficient point cloud processing.
 - For better feature blending, we conduct inter-scale cross-modality interaction by **communicating high-frequency coordinates information**.

Publications (Co-author) (*: equal contribution)

- **PointCLIP V2: Adapting CLIP for Powerful 3D Open-world Learning**
Xiangyang Zhu*, Renrui Zhang*, Bowei He, Ziyao Zeng, Shanghang Zhang, Peng Gao
ICCV 2023 | Link: <https://arxiv.org/abs/2211.11682>
 - We introduce a **realistic shape projection** module, and leverage large-scale language models to **automatically design 3D-semantic prompt**.
 - Our approach significantly surpasses PointCLIP by **+42.90%**, **+40.44%**, and **+28.75% accuracy** on three datasets for zero-shot 3D classification.
- **Binding Touch to Everything: Learning Unified Multimodal Tactile Representations**
Fengyu Yang*, Chao Feng*, Ziyang Chen*, Hyoungseob Park, Daniel Wang, Yiming Dou,
Ziyao Zeng, Xien Chen, Rit Gangopadhyay, Andrew Owens, Alex Wong
CVPR 2024 | Link: <https://cfeng16.github.io/UniTouch/>
 - We align touch embedding with a pre-trained vision-image embedding, using sensor-specific tokens for **multi-sensor** training.
 - We learn a **unified multimodal tactile representation**, enabling touch-LLM, touch-to-image generation, image texture manipulation.
- **ProtoDepth: Unsupervised Continual Depth Completion with Prototypes**
Patrick Rim, Hyoungseob Park, S. Gangopadhyay, Ziyao Zeng, Younjoon Chung, Alex Wong

CVPR 2025 | Link: <https://arxiv.org/abs/2503.12745>

- We present a **prototype-based** approach for **continual learning** of unsupervised depth completion.
- Propose to learn **domain descriptors** that enable the model to select the appropriate prototype set for inference.

- **iQuery: Instruments as Queries for Audio-Visual Sound Separation**

Jiabao Chen, Renrui Zhang, Dongze Lian, Jiaqi Yang, **Ziyao Zeng**, Jianbo Shi

CVPR 2023 | Link: <https://arxiv.org/abs/2212.03814>

- We re-formulate visual-sound separation task and **propose Instrument as Query (iQuery) with a flexible query expansion mechanism**.
- We demonstrate state-of-the-art performance, with **up to 44.2 % improvement** of SDR on MUSIC benchmark.

- **ETA: Energy-based Test-time Adaptation for Depth Completion**

Younjoon Chung, Hyoungseob Park, Patrick Rim, Xiaoran Zhang, Jihe He, **Ziyao Zeng**, Safa Cicek, Byung-Woo Hong, James S. Duncan, Alex Wong

ICCV 2025 | Link: <https://arxiv.org/abs/2508.05989>

- Energy-based test-time adaptation of pretrained depth completion models by **quantifying the likelihood**.
- **Train an energy model** that scores local regions of depth predictions by **exploring the data space with adversarial perturbations**.

- **Enhancing Vision Language Models for 4D Perception**

Seokju Cho, Abhishek Badki, Hang Su, Jindong Jiang, **Ziyao Zeng**, Seungryong Kim, Sifei Liu, Orazio Gallo

Submit to a top-tier conference | Link: to appear on arxiv soon

- We propose **True-Motion Tracking for VLM** to entanglement of camera and object motion, targeting **motion-related scene understanding**.
- We present a QA generation pipeline that generates **large-scale 400K training samples** and a 2.2K-sample benchmark.

- **UniTac: A Unified Multimodal Model for Cross-Sensor Tactile Understanding and Generation**

Jiahang Tu, Fengyu Yang, Chenyang Ma, Xihang Yu, **Ziyao Zeng**, Shaokai Wu, Hanbin Zhao, Zhi Tao, Chao Zhang, Hui Qian

Submit to a top-tier conference | Link: to appear on arxiv soon

- We propose UniTac, the **first Unified multimodal models designed for tactile understanding and generation**.
- UniTac enhances reasoning over physical and cross-sensor information, with a sensor-prior-based sampling that simulates real contact for generation.

- **HOMER: Homography-Based Efficient Multi-view 3D Object Removal**

Jingcheng Ni*, Weiguang Zhao*, Daniel Wang, **Ziyao Zeng**, Chenyu You, Alex Wong, Kaizhu Huang

arXiv technical report, 2025 | Link: <https://arxiv.org/abs/2501.17636>

- **Remove objects** across multi-view images with region-based iteration in a single frame with **homography-based mapping**.
- Present a **3D multi-object removal dataset** with greater object diversity and viewpoint variation than existing datasets.

- **NeuroBind: Towards Unified Multimodal Representations for Neural Signals**

Fengyu Yang*, Chao Feng*, Daniel Wang*, Tianye Wang, **Ziyao Zeng**, Zhiyang Xu,

Hyoungseob Park, Pengliang Ji, Hanbin Zhao, Yuanning Li, Alex Wong

arXiv technical report, 2024 | Link: <https://arxiv.org/abs/2407.14020>

- We propose a **unified neural representation** that aligns multiple neural signal modalities, with pre-trained vision-language embeddings.
- We enhance downstream performance on brain signal classification, retrieval, image reconstruction, and Neuro-LLM.

- **VT-CLIP: Enhancing Vision-Language Models with Visual-guided Texts**

Longtian Qiu, Renrui Zhang, Ziyu Guo, **Ziyao Zeng**, Yafeng Li, Guangnan Zhang

arXiv technical report, 2021 | Link: <https://arxiv.org/abs/2112.02399>

- To adapt CLIP to few-shot classification, we enhance vision-language modeling **via visual-guided texts**.
- Specifically, we **guide text feature to adaptively explore informative regions on the image** and aggregate the visual feature by cross-attention.

Research / Working Experience

- **Nvidia Research** May 2025 – March. 2026
- Research Intern | Work with **Dr. Hang Su**, **Dr. Jindong Jiang**, **Dr. Abhishek Badki**, **Dr. Orazio Gallo**, and **Dr. Sifei Liu**
- **Ziyao Zeng** et al. "Boosting Spatial Understanding in VLMs with Spatial State Tokens." U.S. Patent Application (Pending), NVIDIA Corp., 2026.
- **Shanghai AI Laboratory** Oct. 2022 – March 2023
- Intern Researcher | Project: Visual Grounding, Prototype-based Segmentation
- **GRASP Lab, University of Pennsylvania** April 2021 - Dec. 2022
- Work with **Prof. Jianbo Shi** and **Dr. Renrui Zhang** | Project: DSPoint, iQuery, Reconstruction-based Image Segmentation
- **UISEE AI Lab, UISEE Technology** July 2021 - April 2022
- Intern Algorithm Engineer | Develop point cloud recognition & monocular 3D object detection algorithm for self-driving cars
- **PLUS Lab, ShanghaiTech University** July 2020 - June 2021
- Working with **Prof. Xuming He** and **Zhitong Gao** | Project: Mix-data Anomaly Segmentation, Noisy Labels Classification, Uncertainty Estimation

Service / Tutoring Experience

Reviewer: CVPR (2022, 2025 [**Outstanding Reviewer**], 2026), ICCV (2023, 2025), ECCV (2024, 2026), ICML (2025, 2026), ICLR (2025, 2026), NeurIPS (2024, 2025), ACM MM (2023, 2025), AISTATS (2024, 2025), ICASSP (2024, 2025), BMVC (2026), TCSVT, TMLR
Volunteer of WWF China & Greenpeace; Yale CS PhD '24 Mentor;
ShanghaiTech Freshman Mentor; Intern Lecturer of Shanghai Advanced Research Institute;

Awards / Competitions

Sigma Xi, The Scientific Research Honor Society, Membership Nomination 2025
Merit Student of ShanghaiTech University 2021, 2022
State Grid AI Innovation Competition in Power Dispatch and Control (Top 10%, 24/242) 2021

Skills

Python/PyTorch, C/C++, Unity/C#, Bilingual (Chinese/English: CET6-607, GRE-327+4, TOEFL-103)